



**TRIOS CYBER**  
**LABORATORY PRACTICE 28**

Threat Intelligence / IOC Enrichment

---

**Name: Matthew Josue Portilla Morejon**

# 1. Laboratory Objective and IOC Scope

This laboratory investigated a safe training Indicator of Compromise (IOC) using VirusTotal for external reputation/context enrichment and Wazuh for internal telemetry correlation. The selected IOC was the SHA-256 associated with the EICAR antivirus test file. The workflow compared the external reputation result with the activity visible on the monitored Windows endpoint and documented the context that changes analyst prioritization.

| Evidence item            | Observed value                                                    | Purpose / result                                         |
|--------------------------|-------------------------------------------------------------------|----------------------------------------------------------|
| IOC type                 | SHA-256                                                           | Indicator selected for enrichment                        |
| IOC value                | 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabbf651fd0f | Same indicator used across the laboratory                |
| Reputation source        | VirusTotal                                                        | External reputation and file context                     |
| Observed detection ratio | 63 / 65                                                           | High detection count requiring contextual interpretation |
| Monitored endpoint       | Windows-Agent / ID 002                                            | Internal telemetry source in Wazuh                       |

## 2. VirusTotal IOC Search and Reputation

VirusTotal was opened in Search mode and the SHA-256 indicator was queried directly. The detection view returned 63 detections out of 65 security vendors. Multiple engines labeled the indicator as EICAR, EICAR-Test-File, or similar test-file variants.

The screenshot displays the VirusTotal search results for the SHA-256 indicator `275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabbf651fd0f`. The interface includes a search bar at the top, a file distribution notice, and a community score of 63/65. The file details section shows the file size as 68 B and the last analysis date as 2 minutes ago. The 'DETECTION' tab is active, showing a list of security vendors' analyses. A hand cursor points to the 'Code insights' section, which provides information about the EICAR test string.

**Code insights**

EICAR is a test string used to detect and test antivirus software. It's like a "dummy virus" that triggers an antivirus engine to react, demonstrating its ability to identify and neutralize threats. Here's the key: It's NOT a real virus: EICAR is harmless and cannot infect your computer. It's a standardized test: Almost all antivirus programs are designed to recognize EICAR as a potential threat, ensuring they're working properly. [Show more](#)

**Popular threat label** `virus.eicar/test` **Threat categories** `virus` `trojan` **Family labels** `ecar` `test` `file`

**Security vendors' analysis**

| Vendor    | Detection               | Vendor           | Detection                     |
|-----------|-------------------------|------------------|-------------------------------|
| AhnLab-V3 | Virus/EICAR_Test_File   | Alibaba          | Virus.Win32/EICAR.A           |
| AllCloud  | Engtest:Multi/Eicar     | ALYac            | Misc.Eicar-Test-File          |
| Antiy-AVL | TestFile/Win32.EICAR    | Arcabit          | EICAR-Test-File (not A Virus) |
| Avast     | EICAR Test-NOT Virus!!! | Avast-Mobile     | Eicar                         |
| AVG       | EICAR Test-NOT Virus!!! | Avira (no cloud) | Eicar-Test-Signature          |

<https://www.virustotal.com/gui/search?query=entity%3Afile+tag%3Adirect-cpu-clock-access>

### 3. IOC Metadata Enrichment

The Details view was reviewed to enrich the indicator with file metadata. The captured evidence preserves the SHA-256, other cryptographic hashes, file type information, EICAR identification, file size, and submission/history context. The file size shown in the captured report is 68 B.

**Basic properties**

|           |                                                                                         |
|-----------|-----------------------------------------------------------------------------------------|
| MD5       | 44d88612fea8af36de82e1278abb02f                                                         |
| SHA-1     | 3395856ce81f2b7382dee72602f798b642f14140                                                |
| SHA-256   | 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f                        |
| SSDEEP    | 3a+JraNvgzsVqSwHq9-1JuOgzsko                                                            |
| TLSH      | T141A022003B0EEF2BA20B00200032E8B00808020E2CE00A3820A020B8C83308803EC228                |
| File type | Powershell <span>source</span> <span>powershell</span> <span>ps</span> <span>ps1</span> |
| Magic     | EICAR virus test files                                                                  |
| TrID      | EICAR antivirus test file (100%)                                                        |
| Magika    | POWERSHELL                                                                              |
| File size | 68 B (68 bytes)                                                                         |

**History**

|                        |                         |
|------------------------|-------------------------|
| First Seen In The Wild | 2012-06-14 17:16:14 UTC |
| First Submission       | 2006-05-22 12:42:02 UTC |
| Last Submission        | 2026-09-30 04:01:51 UTC |
| Last Analysis          | 2026-09-30 04:01:51 UTC |

### 4. EICAR Context and Analyst Interpretation

The VirusTotal Code Insights shown in the detection evidence identifies EICAR as a test string used to verify antivirus software and explicitly states that it is not a real virus and cannot infect the computer. This context is critical because the raw detection count alone could otherwise make the indicator appear highly malicious.

**Analyst interpretation:** The 63/65 detection result raises initial suspicion, but the EICAR context materially changes prioritization. In this laboratory the indicator represents a safe security-testing artifact, so it must not be treated as proof of a real malware infection.

### 5. Wazuh Baseline Search

Threat Hunting was opened for Windows-Agent (ID 002). Before a controlled training observation was introduced, the same SHA-256 was searched across the available Wazuh telemetry. The initial query returned no matching results, establishing a baseline in which the IOC had not previously been observed in the selected time range.

Wazuh Threat Hunting - Windows-Agent

Dashboard Events

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f DQL Last 1 week Show dates Refresh

manager.name: trioscyber agent.id: 002 Add filter

No results match your search criteria

## 6. Controlled Training Event Generation

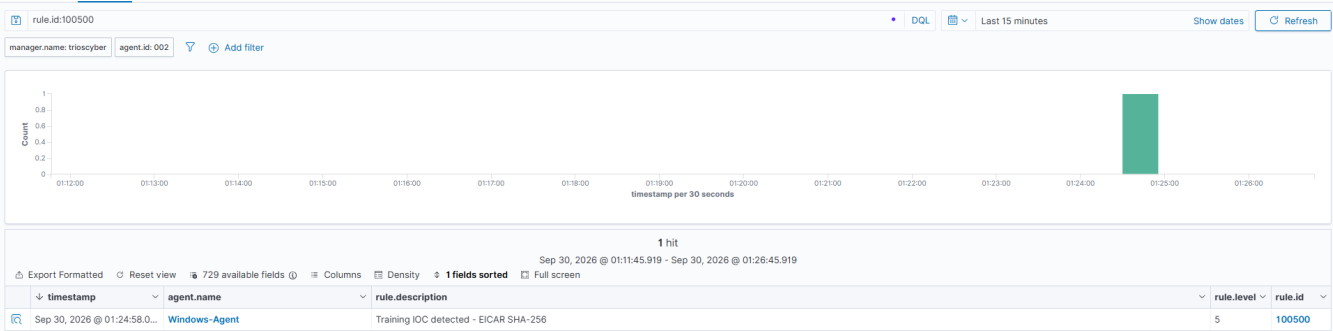
A controlled Windows Application event was created with the source IOCTraining and a message containing IOC\_TEST, the EICAR label, and the selected SHA-256. The command returned a successful event-creation message. This step was intentionally performed for laboratory validation and did not represent a real compromise.

```
matthew@DESKTOP-VOJCDSO C:\Users\matthew>eventcreate /T INFORMATION /ID 100
/L APPLICATION /SO IOCTraining /D "IOC_TEST EICAR SHA256=275a021bbfb6489e54d
471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f SOURCE=TRAINING"

CORRECTO: se ha creado un evento de tipo 'INFORMATION' en el registro 'APPLI
CATION'
con 'IOCTraining' como origen.
```

## 7. Wazuh IOC Detection

After the controlled laboratory observation was introduced, Wazuh Threat Hunting displayed an alert for Windows-Agent. Filtering by rule.id:100500 returned one matching event with the description Training IOC detected - EICAR SHA-256 at level 5. This demonstrates the correlation path from a known training IOC to internal Wazuh evidence.



| Field           | Observed value                        | Meaning                                       |
|-----------------|---------------------------------------|-----------------------------------------------|
| Agent           | Windows-Agent / ID 002                | Endpoint associated with the laboratory event |
| Rule ID / level | 100500 / Level 5                      | Training IOC alert classification             |
| Description     | Training IOC detected - EICAR SHA-256 | IOC-specific laboratory alert                 |
| Hits            | 1                                     | One matching controlled event                 |

## 8. Detailed Wazuh Event Evidence

The Wazuh Document Details view confirms the context of the selected alert. The record identifies Windows-Agent (ID 002), indicator EICAR, IOC type sha256, the full SHA-256 value, source training, location synthetic-lab-event, manager trioscyber, rule 100500 at level 5, and the description Training IOC detected - EICAR SHA-256. The full\_log field also preserves the IOC\_TEST message and the SHA-256.

Document Details

[View surrounding documents](#)

[View single document](#)

TableJSON

|                    |                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------|
| t _index           | wazuh-alerts-4.x-2026.09.30                                                                            |
| t agent.id         | 002                                                                                                    |
| t agent.name       | Windows-Agent                                                                                          |
| data.indicator     | EICAR                                                                                                  |
| data.ioc_type      | sha256                                                                                                 |
| data.ioc_value     | 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f                                       |
| data.source        | training                                                                                               |
| t decoder.name     | json                                                                                                   |
| t full_log         | IOC_TEST EICAR SHA256=275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f SOURCE=TRAINING |
| t id               | 1790749498.100500                                                                                      |
| t input.type       | log                                                                                                    |
| t location         | synthetic-lab-event                                                                                    |
| t manager.name     | trioscyber                                                                                             |
| t rule.description | Training IOC detected - EICAR SHA-256                                                                  |
| # rule.firedtimes  | 1                                                                                                      |
| t rule.groups      | ioc_training, threat_intelligence                                                                      |
| t rule.id          | 100500                                                                                                 |
| # rule.level       | 5                                                                                                      |
| rule.mail          | false                                                                                                  |
| timestamp          | Sep 30, 2026 @ 01:24:58.000                                                                            |

| Field                | Observed value                    |
|----------------------|-----------------------------------|
| data.indicator       | EICAR                             |
| data.ioc_type        | sha256                            |
| data.source          | training                          |
| location             | synthetic-lab-event               |
| rule.id / rule.level | 100500 / 5                        |
| rule.groups          | ioc_training, threat_intelligence |

## 9. Analyst Prioritization and Triage Context

IOC enrichment is useful because reputation alone does not determine incident priority. The same indicator can require different responses depending on whether it is a known test artifact, whether it is actually present in internal telemetry, how it appeared, and whether related activity is expected or unexplained. The following considerations summarize how context would influence analyst prioritization in this laboratory.

| Context                   | Observed / possible condition                               | Effect on prioritization                                    |
|---------------------------|-------------------------------------------------------------|-------------------------------------------------------------|
| External reputation       | 63/65 VirusTotal detections                                 | Raises initial suspicion and justifies investigation        |
| Indicator identity        | EICAR test string; not a real virus                         | Strongly reduces maliciousness assessment in this lab       |
| Initial internal presence | No Wazuh matches before the test                            | No evidence of prior observation in the selected telemetry  |
| Controlled internal match | Rule 100500, level 5, source training                       | Confirms monitoring/correlation path, not a real compromise |
| Asset / source context    | Unexpected appearance on a production asset                 | Would increase priority and require validation              |
| Related behavior          | Unexplained execution, network activity, or repeated alerts | Would increase priority and warrant deeper triage           |

## 10. Validation Summary

| Requirement            | Observed result                                        | Status    |
|------------------------|--------------------------------------------------------|-----------|
| Safe IOC selected      | EICAR SHA-256 used as the training indicator           | Completed |
| External enrichment    | VirusTotal detection, labels, and metadata reviewed    | Verified  |
| Initial Wazuh check    | No matching IOC event before controlled observation    | Verified  |
| Controlled observation | IOC-bearing laboratory event generated                 | Completed |
| Wazuh correlation      | Rule 100500 / Level 5 alert visible for Windows-Agent  | Verified  |
| Prioritization context | Reputation and internal telemetry interpreted together | Completed |

## 11. Conclusions

The Threat Intelligence / IOC Enrichment laboratory was completed by investigating a safe SHA-256 indicator in VirusTotal and correlating the same indicator with Wazuh telemetry. VirusTotal reported a high detection ratio, but its own context identified the indicator as the EICAR antivirus test string rather than a real virus. An initial Wazuh search returned no matches, after which a controlled training observation was introduced and a Wazuh alert was recorded under rule 100500 at level 5. The exercise demonstrates that analyst prioritization should combine external reputation with internal telemetry and operational context instead of relying on detection counts alone.